Why Do Liquid Drops Bounce?

A fourth-year project in the modelling and computation of the hidden gas film in droplet impact

Project brief for Alfie & Jiya

Abstract

Let a raindrop fall onto a windscreen, or two drops collide in a cloud, and something surprising can happen: instead of merging or splashing, they *bounce*, springing apart as if each were an elastic ball. The secret is a film of *gas*, thinner than a wavelength of light, that gets trapped and squeezed between the approaching surfaces. That cushion of air can push back hard enough to reverse the motion before the surfaces ever touch — though whether it survives that long depends on how smooth the surfaces are. Whether a drop bounces, merges, or shatters is decided in this microscopically thin layer — and capturing it in a full simulation takes days of computer time. This project asks you to build a fast, reduced model of the gas film so that its physics can be explored in seconds rather than days, and to use it to predict *when* drops bounce. This document explains what the problem is, what physics you will need beyond the core modules, what has already been done, and how you might start. It is meant as a starting point. If you would like to read into the problem over the summer you are very welcome to — work through it with pencil and paper and attempt the exercises in the boxes as you go — but there is no expectation that you do so before the project formally begins. There are no solutions printed here on purpose: try each exercise, then come and talk them through with us (a solution sheet exists for that conversation).

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1 The phenomenon, and the question

When a liquid drop approaches a solid wall (or another drop) it does not, at first, touch it. The gas in between has to be pushed out of the way, and squeezing gas out of a closing gap takes pressure — more and more of it as the gap narrows. That pressure builds up under the drop, flattens its underside, and can bring the approach to a halt and reverse it while a thin layer of gas is still trapped in the middle. If the film survives, the surfaces never make contact and the drop **bounces** on its own cushion of air.

Whether the film survives that long is not guaranteed, and it is worth being honest about this from the start. On a *solid* surface the trapped film is generally short-lived: any roughness of the solid, or the van der Waals attraction of Part 2, punctures the nanometre-thin gas layer and the liquid wets the surface — the calculations of Mandre, Mani & Brenner (2009) and the experiments of Kolinski *et al.* (2012) show the drop briefly *skating* on a film of air before it ruptures and contact spreads outwards. A true, lasting bounce is far more robust in a *drop–drop* (or drop–liquid) collision, because the two interfaces are themselves smooth and deformable: there is no roughness to seed a puncture, and both surfaces yield to the cushion. This is why clean bouncing is seen most clearly between drops, and why the drop–drop picture below is the cleaner setting for the model. That said, drops *can* bounce off a solid: Kolinski *et al.* (2014) and de Ruiter *et al.* (2015) both observe a full rebound on a flat solid mediated entirely by the air film, even on a perfectly wetting (hydrophilic) surface where the liquid would eventually spread. These are in the minority, though, and they need a *nano-smooth* solid — polished flat enough that no asperity punctures the film before the cushion can act. On an everyday rough surface the film loses this race; between two drops it does not have to.

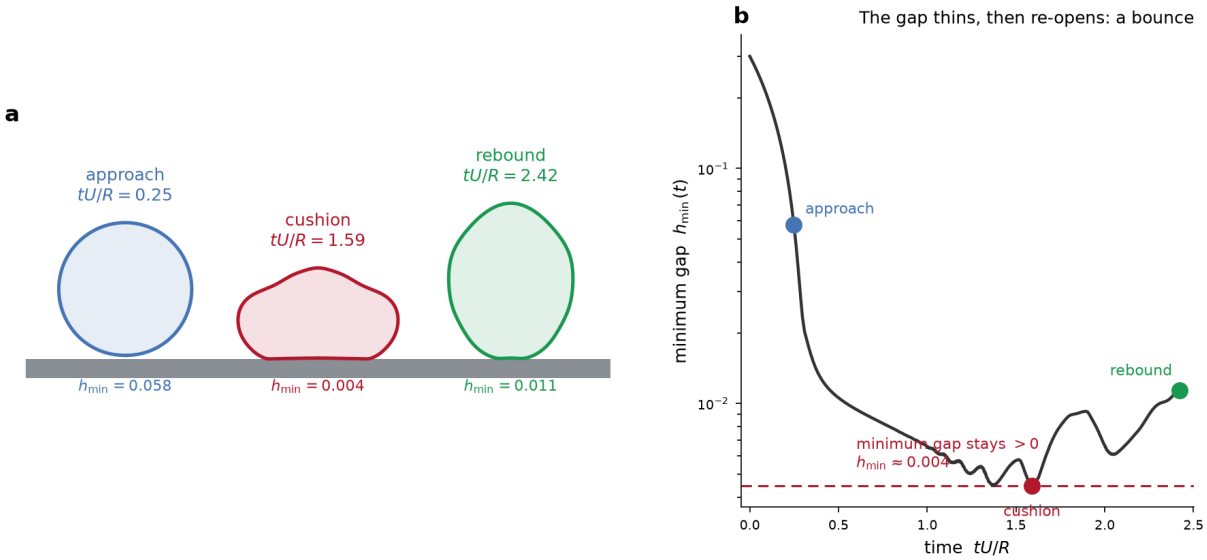


Figure 1: A cushioned bounce computed with a prototype of the kind of solver you will build. **(a)** Snapshots of the drop as it approaches the wall (blue), is flattened against the trapped gas at closest approach (red), and rebounds (green); time is measured in units of R/U. **(b)** The minimum gap between drop and wall over time, on a logarithmic scale. It thins by two orders of magnitude, bottoms out at a few thousandths of the drop radius, and then *re-opens* — the signature of a bounce. In this computation the gap stays positive throughout, so the surfaces never touch; whether a real film lasts that long depends on the smoothness of the surfaces (see text). This is the behaviour the project is about reproducing and understanding.

Whether this actually happens depends, remarkably, on how fast the drop is going. The natural measure of speed is the **Weber number**

$$\mathrm{We} \;=\; \frac{\rho R U^2}{\gamma}, \tag{1}$$

the ratio of inertia (which wants to keep the drop moving and deform it) to surface tension γ (which resists deformation); here ρ is the liquid density, R the drop radius and U the impact speed. Experiments on colliding drops (Figure 2) show a counter-intuitive pattern: at *low* speed the drops merge, at *intermediate* speed they bounce, and at *high* speed they merge again. The bounce lives in a *window* of Weber number, bracketed by contact on both sides. Explaining where the two edges of that window come from — and predicting how they move as you change the liquid, the gas, or the drop size — is the scientific goal.

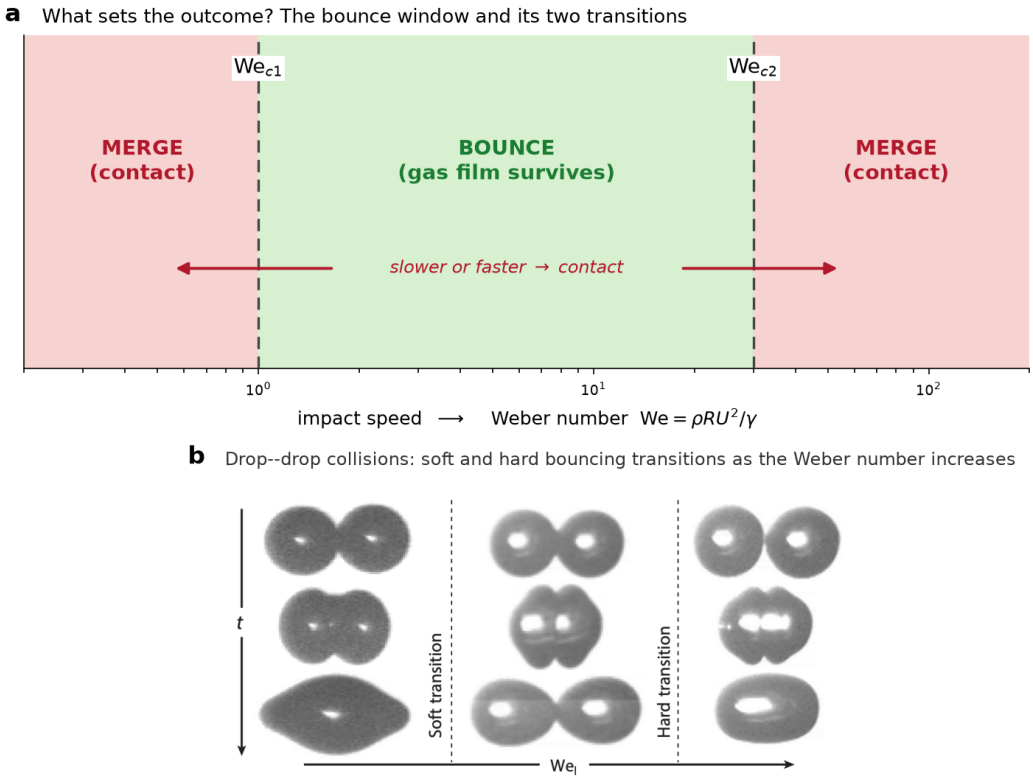


Figure 2: The outcome of a collision as a function of impact speed. **(a)** A **bounce window** sits between two critical Weber numbers We_{c1} and We_{c2}: slow down or speed up out of the window and the gas film fails, the surfaces touch, and the drops merge. Reproducing this map — and understanding what sets We_{c1} and We_{c2} — is the target; the values shown are schematic and the real ones are for you to compute. **(b)** Head-on drop–drop collisions imaged as the Weber number increases, showing the two transitions that bracket the bounce window (reproduced from Sprittles, Annu. Rev. Fluid Mech. **56**, 91 (2024), the review at the top of the reading list, §10 — worth reading first).

1.1 What the project is really about: the film

It is tempting to think you must simulate the whole drop — its shape, its internal flow, its rebound — to answer this. You may not, and the central insight of this project is *why*. Everything interesting happens in the **gas film**: the thin, cushioning layer of gas trapped between the approaching surfaces. The pressure that decides the drop's fate is generated *there*, in a region whose radial extent is a small fraction of the drop radius and whose thickness is smaller still.

This is exactly the point of view taken by **Hicks & Purvis** (2010), whose approach this project follows. They do *not* solve for the whole drop. They focus on the **film region** alone, and they make one decisive simplifying assumption: the underside of the drop *begins* as a gentle **parabola**,

$$z_{\mathrm{drop}}(r, t=0) \;\approx\; h_0 \;+\; \frac{r^2}{2R}, \qquad r \text{ small}, \tag{2}$$

where h_0 is the initial gap on the axis and R is the (local) radius of curvature of the drop bottom. This is an *initial* (and far-field) condition, *not* a shape imposed for all time: once the film pressure builds up, the underside deforms — it flattens and then **dimples** — and the whole point of the method (§4) is to compute that evolving shape. The parabola is where the drop *starts*; the film decides where it goes. *This project is about the film part.* We will treat the gas film as the object of study, with the drop surface entering only through its evolving shape (2) and the speed at

which it descends. The simplest version of the model (the kinematic benchmark, Exercise 6) does freeze the underside as a rigid descending parabola; the fuller method (§4) relaxes that and lets the film shape the drop.

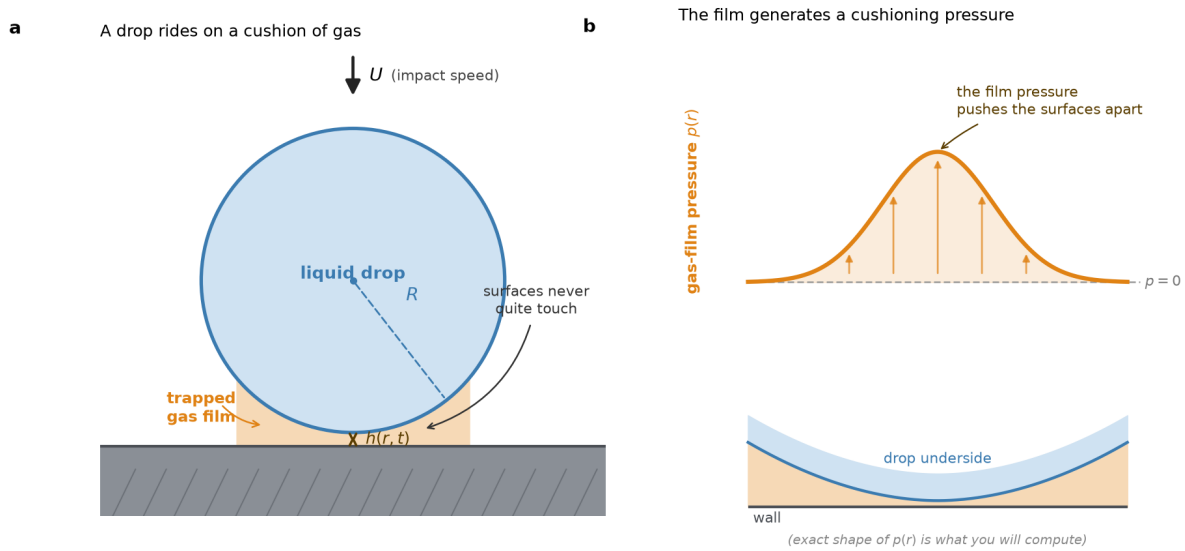


Figure 3: **(a)** The geometry. A drop of radius R descends at speed U onto a wall; a film of gas of thickness $h(r,t)$ is trapped in between. Following Hicks & Purvis, we study *this film*, taking the drop underside to *start* as a parabola (2). **(b)** The idea of the cushion, schematically: gas squeezed out of the closing gap builds up a pressure that pushes back on the underside and resists the approach. The precise shape of $p(r)$ — whether it peaks on the axis or off it, and whether it develops a region of suction — is exactly what the film model predicts and what you will compute; it is deliberately *not* committed to here.

Part 1

The cushion holds: the gas as a simple viscous film

This first part is about the gas film in its simplest honest form: the gas is a simple viscous fluid governed by the standard lubrication (Reynolds) equation, and there are no attractive forces between the surfaces. We treat the gas as a continuous fluid throughout this project — that never changes; what Part 3 later adds is a small rarefaction (slip) correction to this same continuum description, not a different kind of model. Under those assumptions the film generates a purely *repulsive* pressure that can hold the drop up — the cushion holds. Your goal in Part 1 is to derive the equation that governs the film pressure, understand the scaling that controls it, and turn it into a small solver that reproduces a bounce like Figure 1. We hold back van der Waals attraction (Part 2) and gas-kinetic effects (Part 3) until this simplest cushion is under control.

Part 1 has two halves: **Part 1A** is pencil-and-paper (the lubrication equation for the film and the scaling that governs it); **Part 1B** turns that equation into code and tests it against a benchmark. The boundary-integral method of Hicks & Purvis — how the film pressure is allowed to shape the drop underside, still without leaving the film region — is set out in §4.

Part 1A *pencil and paper:* **the lubrication film**

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2 Lubrication in the gas film

The film is thin and wide: its thickness h is tiny compared with the radial distance r over which it varies. That disparity of scales is what makes the problem tractable, exactly as it does for a thin liquid film, and it leads to the same reduction — **lubrication theory**. Across the thin gap the pressure is essentially uniform, the flow is a Poiseuille profile driven by the radial pressure gradient, and mass conservation for the gas being squeezed out closes the problem. The result is the **Reynolds equation** for the film.

As a starting point we recommend working throughout in **planar (two-dimensional)** geometry: treat the film as a strip, with the gap $h(x,t)$ and pressure $p(x,t)$ depending on a single in-plane coordinate x. This is the simplest honest setting, it already contains all of the physics of the cushion, and it is where you should begin — the axisymmetric case (the real spherical-drop geometry) is a change of geometry we build up to later (§4.2 and Appendix A). For an incompressible gas of viscosity μ_g, the planar Reynolds equation is

$$\frac{\partial}{\partial x}\left(\frac{h^3}{12\,\mu_g}\frac{\partial p}{\partial x}\right) = \frac{\partial h}{\partial t}. \tag{3}$$

The axisymmetric form that Hicks & Purvis actually solve simply restores the radial geometry, replacing ∂_x by the radial divergence with its $1/r$ factor,

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\,\frac{h^3}{12\,\mu_g}\frac{\partial p}{\partial r}\right) = \frac{\partial h}{\partial t}, \tag{4}$$

where p is the gas pressure (measured relative to the ambient) and h is the gap. In the axisymmetric case we are using a **cylindrical polar coordinate system** (r,θ,z) aligned with the axis of symmetry: z points along the line of impact, r is the distance from that axis, and nothing depends on the azimuthal angle θ, so the film is described by $h(r,t)$ and $p(r,t)$ alone. The $1/r$ factors in (4) are exactly the ones the radial divergence carries in polar coordinates; they are absent in the planar strip (3), where x is an ordinary Cartesian coordinate. Figure 4 shows the two geometries side by side. In either geometry the right-hand side is the rate at which the gap closes, and the left-hand side is the divergence of the gas flux squeezed out sideways. The cube h^3 is the crucial nonlinearity: as the gap narrows the film becomes dramatically harder to drain, so the pressure needed to keep squeezing gas out shoots up. That is the physical origin of the cushion.

A note on notation. We attach a subscript to *gas* properties — the gas viscosity μ_g, and later the gas mean free path λ and the gas-based Knudsen and Stokes numbers — but leave *liquid* properties unsubscripted (the liquid density ρ, surface tension γ). The gas is the special, thin, hard-working phase here, so it is the one we mark; the liquid is the "default".

Exercise 1. Deriving the Reynolds equation

Starting from the Stokes equations for the gas in the thin gap, assume the lubrication scalings ($h \ll r$, pressure uniform across the gap, radial flow much larger than transverse). Integrate the radial momentum balance across the gap with no-slip on the drop underside and on the wall to obtain the Poiseuille flux $q(r,t) = -(h^3/12\mu_g)\,\partial_r p$ per unit circumference. Then impose conservation of gas mass in an annular control volume to arrive at (4). Where exactly did you use $h \ll r$? Do the planar case first — a flat slab control volume gives (3) with no $1/r$ factors — and only then repeat the mass balance on an annulus to recover the axisymmetric form. That is the order in which we suggest you build everything.

To make sense of (4) it helps to non-dimensionalise. Scale lengths with the drop radius R, speeds with the impact speed U, time with R/U, and pressure with the inertial scale ρU^2. A

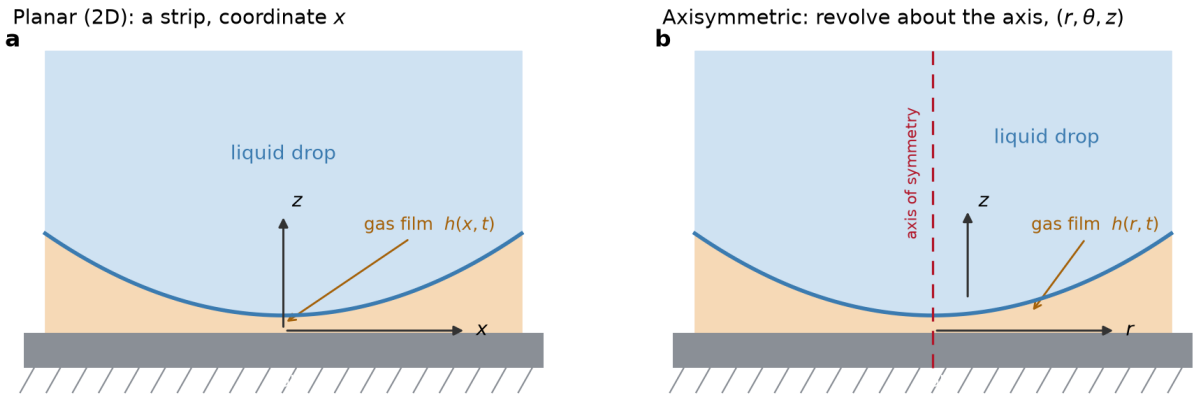


Figure 4: The film region in the two geometries. **(a)** Planar (2D): the film is a strip and everything depends on a single Cartesian coordinate x, so the Reynolds equation (3) has no $1/r$ factors. This is where we recommend you start. **(b)** Axisymmetric: we use a cylindrical polar system (r, θ, z) about the line of impact, the picture is obtained by revolving the profile about the dashed axis, and the radial divergence supplies the $1/r$ factors of (4). In both, the gap h is the thickness of the (orange) gas film between the drop underside and the wall.

single dimensionless group then controls the film,

$$\mathrm{St} \;=\; \frac{\rho R U}{\mu_g}, \tag{5}$$

the **Stokes number**, comparing liquid inertia with gas viscosity. In these units the (incompressible) Reynolds equation becomes

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\,h^3\,\frac{\partial p}{\partial r}\right) \;=\; \frac{12}{\mathrm{St}}\,\frac{\partial h}{\partial t}, \tag{6}$$

or, in the planar geometry we suggest you start with, $\partial_x(h^3\,\partial_x p) = (12/\mathrm{St})\,\partial_t h$. So St is the knob that sets how stiff the gas cushion is. Large St (fast, big drops; thin gas) means a small right-hand side — the film drains easily until the gap is very small, and the cushioning only bites at extreme thinness.

Exercise 2. Non-dimensionalising the film, and the meaning of St

Carry out the non-dimensionalisation that turns (4) into (6), and confirm that $\mathrm{St} = \rho R U/\mu_g$ is the only dimensionless group that appears. Estimate St for a millimetre water drop hitting a wall at $1\,\mathrm{m\,s^{-1}}$ in air ($\mu_g \approx 1.8\times10^{-5}\,\mathrm{Pa\,s}$). You should get a large number — explain in words what that implies about how thin the gap must become before the cushion matters.

2.1 The parabolic drop surface and the squeeze-flow pressure

Now use the Hicks & Purvis picture, and stay planar. Take the drop underside to descend as a rigid parabola, $h(x,t) = h_0(t) + x^2/2$ (radius of curvature scaled to $R = 1$), closing at speed $V = -\dot{h}_0 > 0$. Then $\partial_t h = \dot{h}_0 = -V$ is *uniform* in x, and the planar equation (3) can be integrated once in closed form. Imposing symmetry at $x = 0$ and $p \to 0$ at the outer edge gives a pressure

$$p(x,t) \;=\; 6\,\mu_g\,V\,\frac{1}{h(x)^2} \;+\; \text{const}, \tag{7}$$

the classic **squeeze-film** result: the pressure is largest where the gap is thinnest, and it scales like $1/h^2$. This single formula already tells you most of the story — the force under the drop

grows without bound as $h_0 \to 0$ if nothing intervenes, which is precisely why the drop can be turned around. The axisymmetric version of the same calculation, integrating (4) on the parabola $h = h_0 + r^2/2$, gives $p = 3\,\mu_g V/h^2$: the same $1/h^2$ law, with half the prefactor because the gas now escapes radially in two directions rather than one. Note that the factor 12 in the flux law is common to both geometries — it comes from the Poiseuille profile *across* the gap, which knows nothing about the in-plane geometry — and only the geometric prefactor (6 planar, 3 axisymmetric) changes.

Exercise 3. The squeeze-film pressure and the cushioning force

Working in planar geometry, for the parabolic gap $h = h_0 + x^2/2$ closing at speed V, solve (3) to obtain the pressure (7) explicitly (fixing the constant so that $p \to 0$ as $x \to \infty$). Then integrate the pressure over the underside to find the total upward force per unit length $F(h_0) = \int_{-\infty}^{\infty} p\,\mathrm{d}x$. How does F diverge as $h_0 \to 0$? This divergence is the mathematical statement of "the cushion holds". Then repeat the calculation in the axisymmetric geometry ($F = \int_0^{\infty} p\,2\pi r\,\mathrm{d}r$) and confirm you get the $3\mu_g V/h^2$ prefactor.

There is one more essential feature the simple parabola misses but your solver will capture: the underside does not stay parabolic. The build-up of film pressure decelerates the *centre* of the underside first, so that while the middle is held up the surface continues to descend around it; the thinnest point migrates off-centre and the gap turns into a **dimple** — a trapped pocket of gas rimmed by a thin **barrier ring** where the film is thinnest. The minimum gap therefore sits away from the centre, and the thickness of that dimple obeys a scaling with the Stokes number,

$$h_{\mathrm{dimple}} \sim \mathrm{St}^{-2/3}, \tag{8}$$

one of the cleanest quantitative predictions of the theory and an excellent target for your code to reproduce. This dimple is not just a modelling artefact: experiments show that when a drop impacts a solid it leaves behind a small *air bubble* — the remnant of exactly this trapped central pocket — and both the entrapped-bubble size and its dependence on impact speed have been measured directly (Thoroddsen *et al.*, 2005; Bouwhuis *et al.*, 2012; reading list, §10). Their scalings are the experimental counterpart of (8).

Exercise 4. The dimple scaling (harder)

By balancing the terms in (6) at the moment the film pressure first becomes large enough to resist the approach — i.e. when the film force becomes comparable to the drop's inertia — estimate how the trapped-dimple thickness scales with St, and recover (8). This is a scaling argument, not an exact solution: identify the dominant balance, not the prefactor.

Part 1B *at the keyboard:* **a first solver**

3 Turning the film into code

The analytic results above are limiting cases; the real film equation has to be solved numerically because the gap develops the dimple and the pressure peak migrates. The good news is that it is a single one-dimensional PDE, and a compact, robust discretisation is well within reach.

Start in the planar geometry: solve $\partial_x(h^3\,\partial_x p) = (12/\mathrm{St})\,\partial_t h$ on a grid in x. This is the cleanest first build — no $1/r$ factors, symmetric about $x = 0$ — and everything below carries over to the radial equation (6) with only the geometric factors changed. The recommended approach is a **finite-volume** scheme on a grid $x_0 < x_1 < \cdots < x_M$. Writing the conductance $K = h^3$, the flux through the face between cells i and $i{+}1$ is $-K_{i+1/2}(p_{i+1} - p_i)/\Delta x$, and demanding that

the net flux out of each cell balance the local $(12/\mathrm{St})\,\partial_t h$ gives a tridiagonal linear system for the nodal pressures p_i. Symmetry ($\partial_x p = 0$) closes the centreline; $p = 0$ closes the outer edge. This is second-order accurate and can be checked exactly against the squeeze-film solution (7). Promoting the code to the axisymmetric equation (6) is then a one-line change: restore the $r_{i+1/2}$ weighting on each face and the $\partial_r p = 0$ condition on the axis.

Exercise 5. A finite-volume Reynolds solver

Working in planar geometry to begin with, implement the finite-volume discretisation on a grid and solve the resulting tridiagonal system (Python's `scipy.linalg.solve_banded` is ideal). Validate it in two independent ways: (i) against the analytic squeeze-film pressure (7) for a fixed parabolic gap closing at constant speed; (ii) with a manufactured solution — pick a smooth $p(x)$, substitute to find the source it requires, and confirm your code recovers it at second-order as you refine the grid. *Always check a computation against something you can compute independently.* Once this works, switch on the radial weighting and repeat the checks in the axisymmetric geometry.

With the pressure solver in hand, the **kinematic benchmark** is to march the gap forward in time. Take the drop underside to be the descending parabola (2) — so its shape is prescribed *kinematically*, not yet computed from the liquid dynamics — let the film pressure it generates decelerate its descent through Newton's second law for the drop's vertical motion, and watch what happens to the minimum gap. If your solver is right you will see the approach slow, the dimple form, and (for the right range of parameters) the gap stop shrinking and re-open: a bounce. That is the first milestone, and it already contains the essential physics of Figure 1 without any of the harder machinery below.

It is worth being clear about what this benchmark is, physically: because the underside shape is frozen as a rigid parabola, the drop enters the problem only as a *rigid body* — a ball of effective mass m_{eff} whose one degree of freedom is the height $h_0(t)$ of its lowest point. All the deformation physics is switched off; the only thing that can turn the ball around is the cushioning force of the gas film. So this exercise is, quite literally, a rigid ball bouncing on a cushion of air: the film is the spring, and the question is whether that spring is stiff enough to reverse the ball before the gap closes. The genuinely deformable drop comes later (§4); getting the rigid-ball bounce right first isolates the cushioning mechanism from the harder coupling.

Exercise 6. The kinematic benchmark: a rigid ball on a gas cushion

Model the drop as a rigid ball: its underside is the fixed parabola (2) and its only unknown is the axial gap $h_0(t)$. Couple your Reynolds solver to a single ordinary differential equation for that gap, $m_{\mathrm{eff}}\,\ddot{h}_0 = -F(h_0, \dot{h}_0)$, where F is the film force from Exercise 3 (now evaluated numerically on the evolving gap) and m_{eff} is the effective mass of the ball. March it in time with a standard explicit integrator. For an impact speed in the bouncing range, show that the minimum gap reaches a positive minimum and then grows again — the ball has bounced off the air without ever touching the wall. Plot $h_{\min}(t)$ and compare its shape with Figure 1b. As a check on the physics, confirm that a stiffer cushion (larger St) or a lighter ball rebounds more readily.

4 Letting the film shape the drop: the Hicks & Purvis method

The kinematic benchmark of Exercise 6 *prescribes* the underside as a rigid parabola descending at a fixed speed. The genuine model of Hicks & Purvis (2010) does one thing more: it lets the film pressure deform the underside, and closes the loop by asking the liquid how it responds. Crucially, this is still a *film-region* calculation — there is no mesh of the whole drop and no

velocity potential to solve for. It rests on a single observation about the liquid in the impact zone, and it is well worth understanding, because it is the method the project is built on.

In the thin local region near closest approach the liquid moves fast and is barely deflected, so to leading order it is *inviscid* and its acceleration balances the pressure gradient, $\partial \mathbf{u}/\partial t = -\nabla p$ with $\nabla \cdot \mathbf{u} = 0$. Taking the divergence shows the liquid pressure is **harmonic**,

$$\nabla^2 p \;=\; 0 \qquad \text{in the liquid,} \tag{9}$$

and this is what makes the drop tractable without solving for it. A harmonic field in the half-space above the wall is fixed by its values on the boundary, so the whole liquid problem collapses onto the interface.

To turn "fixed by its boundary values" into a formula we use a **Green's function**. You have already met the ingredient in another guise: in your methods course (MA3D1) you saw how a field responds to a *point* forcing — a delta function — and how the response to any distributed forcing is then built by adding up those point responses. A Green's function is exactly that: $G(\mathbf{x}, \mathbf{x}')$ is the harmonic field at $\mathbf{x}$ produced by a unit point source at $\mathbf{x}'$, and a general harmonic field can be reconstructed as an integral of G against its boundary data. The one new feature here is the wall. We need a G that not only satisfies $\nabla^2 G = 0$ but also respects the wall at $z = 0$, and the device that arranges this is the **method of images**: alongside the real source in the liquid we place a fictitious *image sink* at the mirror-image point reflected below the wall. On the plane $z = 0$ the source and its image cancel exactly, so their combined field meets the wall condition there automatically, while above the wall it still solves Laplace's equation. The image is not a physical object — it is the bookkeeping trick that bakes the wall into G once and for all, so the wall never has to be imposed again. (You check both properties in Exercise 7.)

With that Green's function the harmonic problem for the liquid pressure reduces to an integral over the interface alone — no interior mesh and no velocity field to track, which is what makes the method so cheap. We set it up, as throughout, in the **planar** geometry first: the liquid fills the half-plane $z > 0$ above the wall and depends on a single in-plane coordinate x. After using the kinematic condition on the underside, the boundary relation between the interface acceleration and the interfacial pressure is

$$\frac{\partial^2 f}{\partial t^2}(x,t) \;=\; \frac{1}{\pi}\,\mathrm{PV}\!\int \frac{p_{x'}(x',t)}{x'-x}\,\mathrm{d}x', \tag{10}$$

where $f(x,t)$ is the height of the underside above the wall, $p_{x'}$ is the tangential pressure gradient on the wall, and the integral is a Cauchy principal value. This is the **boundary integral**: a single *one-dimensional* integral over the film's footprint, with no interior unknowns at all. It is a **Hilbert transform** of the interfacial pressure gradient — exactly the relation used by Mani, Mandre & Brenner (2010) in their two-dimensional analysis of the same problem. The axisymmetric form, a single radial integral, is given in §4.2.

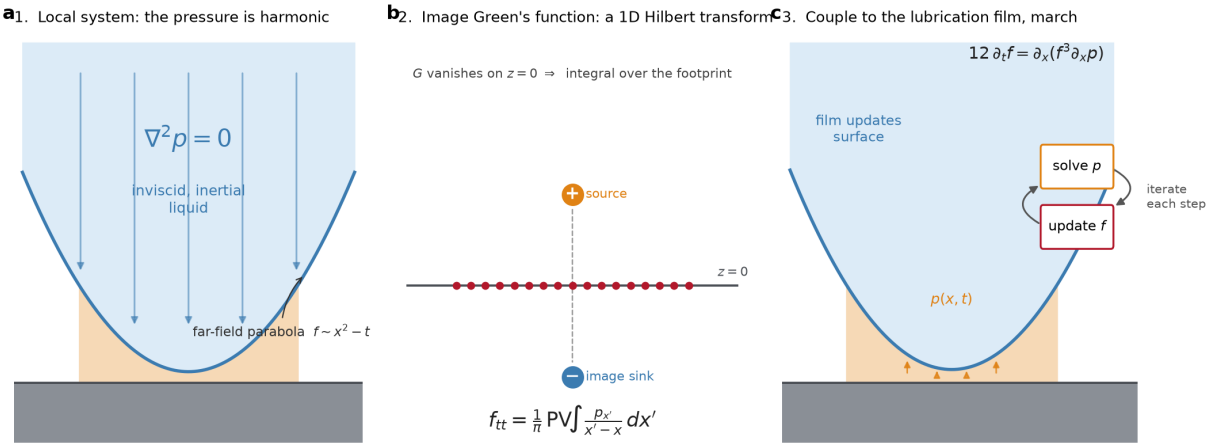


Figure 5: The Hicks & Purvis film-region method (incompressible), drawn in the planar geometry you build first. **(1)** In the thin impact region the liquid is inviscid and inertial, so its pressure is harmonic, $\nabla^2 p = 0$; the underside descends onto the wall with the far-field parabola $f \sim x^2 - t$. **(2)** Because a Green's function that vanishes on the wall can be built from a source and its *image sink* below the wall, the harmonic problem reduces to an integral over the film's footprint on the wall — relation (10) between the interface acceleration f_{tt} and the interfacial pressure. **(3)** That pressure is supplied by the lubrication film (11); the two are solved together and marched in time. The wall never enters the liquid problem directly — it bounds the *gas*, and the gas transmits its effect to the liquid only through the interfacial pressure p.

The second half of the coupling is the film itself. Written in the same local variables, and still planar, the lubrication equation (3) for the gas between the underside and the wall is

$$12\,\frac{\partial f}{\partial t} \;=\; \frac{\partial}{\partial x}\left(f^3\,\frac{\partial p}{\partial x}\right), \tag{11}$$

now with f (the drop underside) playing the role of the gap. Equations (10) and (11) are a closed pair: the film equation gives the pressure for a given gap, and the boundary integral gives the new gap for a given pressure. Solved together — iterated to convergence at each step and marched forward in time (§4.1) — they reproduce the cushioned approach and the dimple without ever prescribing the drop shape. This is the model the project extends in Parts 2 and 3; the parabola is only its far-field condition, not a permanent assumption. Everything here is written in the planar geometry you build first; the axisymmetric equivalents are collected in §4.2 and Appendix A.

4.1 The numerical scheme

The recipe is worth stating, because it is exactly what you will implement. Do it in planar geometry first. On a grid over a truncated window (say $x \in [-6, 6]$), each time step is a **two-step iteration**. First, holding the gap fixed, the film equation (11) is discretised with a three-point stencil for $\partial_x(f^3 \partial_x p)$ and a backward difference in time, and the resulting tridiagonal system is solved for the pressure. Second, that pressure is fed into a *second-order* discretisation of the boundary integral (10) — the principal-value integrand is desingularised by subtracting its value at the target point, so the $1/(x' - x)$ kernel integrates cleanly — to advance the underside f. The two steps are repeated until the pressure and surface stop changing (an r.m.s. tolerance of $\sim 10^{-8}$), typically a handful of iterations early on rising to a few hundred as the gap narrows. A small time step ($\Delta t \approx 0.02$) and release from far away ($t = -8$, undeformed, zero film pressure) complete the scheme. The axisymmetric target you build up to (§4.2) keeps every one of these ingredients; it replaces the x-grid by a radial grid in r, the ∂_x stencil by its $1/r$-weighted radial

form, and the Hilbert kernel by the elliptic-integral kernel of (13). Hicks & Purvis solve exactly this on a polar grid (a few hundred radial points) and cross-check it against a full Cartesian run — but that is the second build, not where you begin.

Exercise 7. The image Green's function

Verify that $G(\mathbf{x}', \mathbf{x}) \propto |\mathbf{x} - \mathbf{x}'|^{-1} - |\mathbf{x} - \mathbf{x}^*|^{-1}$, where $\mathbf{x}^*$ is $\mathbf{x}$ reflected in the wall $z = 0$, satisfies $\nabla^2 G = 0$ away from the source and vanishes on $z = 0$. Explain in one sentence why that boundary condition is exactly what lets the interior liquid problem be replaced by an integral over the wall footprint, as in (10).

Exercise 8. Desingularising the boundary integral

Working with the planar boundary integral (10), the kernel $1/(x' - x)$ diverges as $x' \to x$, which a naive quadrature cannot integrate. Show that subtracting and adding the pressure gradient *at the target point* x — i.e. replacing $p_{x'}(x')$ by $p_{x'}(x') - p_x(x)$ in the numerator — removes the singularity, because the new numerator vanishes as $x' \to x$ and cancels the $1/(x' - x)$ pole (the omitted term integrates to zero in principal value). This is the step that makes the scheme above work. The same idea carries over to the axisymmetric kernel (13).

4.2 From planar to axisymmetric

Everything above — the harmonic pressure, the boundary integral, the lubrication film, and the two-step iteration — has been set up in the planar (two-dimensional) geometry, where the drop is an infinite cylinder rather than a sphere. That is deliberate, and it is where you should build and validate the coupled solver first: the film equation is the one-dimensional form (11), and the boundary integral is the one-dimensional Hilbert transform (10) — both far simpler to discretise and to debug, while carrying all of the physics (cushion, dimple, bounce).

The real impact of a spherical drop onto a flat wall is *axisymmetric*: the gap and the pressure depend only on the distance r from the axis of impact, not on the azimuthal angle. That — and not the fully general three-dimensional case — is the target for this project, and reaching it is the goal once the planar solver works. Hicks & Purvis confirm the flow is axisymmetric and, as an accuracy check, run their scheme both on a full Cartesian mesh and on a polar (axisymmetric) grid; the polar version is the one we build up to. Promoting the planar solver is a change of geometry, not of method. The film equation regains its $1/r$ factor,

$$12\,\frac{\partial f}{\partial t} \;=\; \frac{1}{r}\frac{\partial}{\partial r}\left(r\,f^3\,\frac{\partial p}{\partial r}\right), \tag{12}$$

and the boundary integral (10) becomes a single *radial* integral over the footprint. Starting from the surface relation of Hicks & Purvis and doing the azimuthal integration analytically — legitimate precisely because f and p no longer depend on angle — leaves a one-dimensional integral in r,

$$\frac{\partial^2 f}{\partial t^2}(r,t) \;=\; \int_0^\infty \mathcal{K}(r,r')\,\frac{\partial p}{\partial r'}(r',t)\,\mathrm{d}r', \tag{13}$$

whose kernel $\mathcal{K}(r,r')$ is fixed once and for all and is written in terms of complete elliptic integrals. This is the correct axisymmetric counterpart of the planar Hilbert transform (10): still a single integral, now in r, *not* the double integral over x and y that a fully three-dimensional drop would demand. The two-step iteration and the desingularisation carry over unchanged; what changes is the geometric bookkeeping — the radial face-weighting in the film solve, and the elliptic-integral kernel of (13). The explicit kernel and how it is desingularised are collected in Appendix A so they do not clutter the first build.

Part 2

Adding attraction: van der Waals

Part 1 gave a film that only ever pushes the surfaces apart. But at very small separations — tens of nanometres and below — the surfaces begin to *attract* each other through **van der Waals** forces, the same intermolecular attraction that makes thin liquid films rupture. This attraction works against the cushion: it pulls the surfaces together and can defeat the gas film that would otherwise hold them apart. Part 2 adds it to the model and asks how it moves the merge/bounce boundary.

5 The disjoining pressure

The van der Waals attraction between the drop surface and the wall (or the other drop), acting across the thin gas gap, is represented in the same way as for a thin liquid film: by a **disjoining pressure** $\Pi(h)$ added to the film pressure. For non-retarded van der Waals forces across a gap of thickness h,

$$\Pi(h) \;=\; \frac{A}{h^3}, \tag{14}$$

where A is a constant set by the **Hamaker constant** of the material combination (it carries the sign: attraction between the approaching surfaces corresponds to a pressure that *assists* closure). The film now responds to the *total* normal stress — the lubrication pressure of Part 1 plus this disjoining term — so the effective pressure driving the gas film becomes $p + \Pi(h)$, and the same Reynolds equation (6) is solved with that combined forcing.

Exercise 9. Where van der Waals starts to matter

The disjoining pressure $\Pi = A/h^3$ grows even faster as $h \to 0$ than the squeeze-film pressure $p \sim 1/h^2$ of Exercise 3. Using a typical Hamaker constant ($A \sim 10^{-20}$ J), estimate the gap h at which the van der Waals attraction becomes comparable to the lubrication pressure that is holding the drop up. Compare it with the dimple thickness (8). For which impact speeds (Stokes numbers) does the film get thin enough for van der Waals to intervene before the drop rebounds?

6 The lower edge of the bounce window

Adding (14) to your solver is, as with the thin-film problem, almost literally one line of code — but the consequences are dramatic. Once the dimple thins to the range where $\Pi(h)$ is significant, the attraction can win: the barrier ring is pulled into contact, the gas film ruptures, and the drops merge instead of bouncing. This is a mechanism for the *lower* critical Weber number We_{c1} in Figure 2: too slow, and the film lingers thin for long enough that van der Waals bridges the gap.

Exercise 10. Does the drop still bounce?

Repeat your kinematic benchmark (Exercise 6) with the disjoining pressure (14) switched on. For a fixed impact speed, increase A from zero and find the value at which the outcome flips from bounce (gap re-opens) to merge (minimum gap collapses to the contact scale). Then hold A fixed and vary the impact speed: locate the critical Weber number below which the drop merges. How does We_{c1} depend on A?

Part 3

The nanoscale finish: gas-kinetic effects

There is a subtlety hiding in Part 1. The Reynolds equation (4) treats the gas as a continuous fluid — but when the gap thins to tens of nanometres it becomes comparable to the distance a gas molecule travels between collisions, the **mean free path** λ (about 70 nm for air at room conditions). At that point the gas is no longer a simple continuum, it drains more easily than the continuum theory predicts, and the cushion is *softer* than Part 1 says. Part 3 builds this correction in and asks how it shifts the whole bounce map. This is the frontier of the project.

7 The Knudsen number and slip

The relevant measure is the **Knudsen number**

$$\mathrm{Kn} \;=\; \frac{\lambda}{h}, \tag{15}$$

the ratio of the mean free path to the gap. When $\mathrm{Kn} \ll 1$ the continuum Reynolds equation holds; when Kn becomes order one — which it does in the thinnest part of the dimple — the no-slip condition on the surfaces fails and the gas slips. The standard way to capture this without abandoning lubrication theory is to keep the Reynolds equation but multiply the film conductance by a Kn-dependent correction factor,

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\, h^3\, Q(\mathrm{Kn})\, \frac{\partial p}{\partial r} \right) \;=\; \frac{12}{\mathrm{St}}\,\frac{\partial h}{\partial t}, \qquad Q(\mathrm{Kn}) = 1 + \alpha\,\mathrm{Kn} + \dots, \tag{16}$$

where Q increases with Kn (the gas conducts *better* than continuum), and the leading correction is a slip term linear in Kn. More accurate forms interpolate all the way to the free-molecular limit; part of the project is deciding how much fidelity the physics of the bounce actually needs.

Exercise 11. Estimating the Knudsen correction

For the dimple thickness (8), estimate the Knudsen number in the thinnest part of the film for a millimetre water drop in air at $1\,\mathrm{m\,s^{-1}}$. Is the continuum Reynolds equation of Part 1 actually valid there? Then linearise (16) for small Kn and argue physically why a gas that slips makes the cushion *weaker* (easier to drain) rather than stronger.

8 How the bounce map shifts

Because gas-kinetic effects make the film drain more easily, they let the gap get thinner before the cushion can turn the drop around. A thinner gap means van der Waals (Part 2) is more likely to bridge it — so the gas-kinetic correction and the disjoining pressure work together, and the net effect on the bounce window is a genuine prediction of the coupled model rather than something you can guess. In particular the *upper* critical Weber number We_{c2} — fast drops that merge — is sensitive to how thin the film is allowed to get, and hence to Kn.

Exercise 12. The full map (open-ended)

With both the disjoining pressure (Part 2) and the gas-kinetic correction (16) in your solver, map the outcome (bounce vs merge) over a range of Weber numbers and reconstruct the bounce window of Figure 2. How do We_{c1} and We_{c2} shift when you turn the Knudsen correction on and off? This is a research-level question: there is no printed answer, and the comparison with the experiments in the reading list is exactly the kind of result the project

is aiming to produce.

9 A suggested plan

The project has a natural spine — build the continuum film, add attraction, add gas kinetics — and a lot of freedom in how far you push each stage. What follows is a suggested ordering, not a contract; how far into the later phases you get is up to you and your supervisors.

Phase 1 — The continuum cushion (Part 1). Build and validate the finite-volume Reynolds solver (Exercise 5) against the analytic squeeze-film pressure, then couple it to the drop's vertical motion and reproduce a kinematic bounce (Exercise 6) — the kinematic benchmark. Confirm the dimple forms and, if you can, check the $\mathrm{St}^{-2/3}$ scaling (8). This is your first validation and the template for everything after: *always check a computation against something you can compute independently.*

Phase 2 — Van der Waals and the merge threshold (Part 2). Add the disjoining pressure (14) — one line — and find the lower edge of the bounce window: the critical impact speed below which the film ruptures and the drops merge (Exercise 10). Map how We_{c1} depends on the Hamaker constant.

Phase 3 — Gas-kinetic effects (Part 3). Introduce the Knudsen correction (16) and study how it softens the cushion and shifts the map (Exercises 11, 12). This is the genuinely novel part, and where your two individual projects can diverge: one of you might push the kinetic model toward the free-molecular limit, the other might focus on the van der Waals rupture dynamics, so that between you the whole bounce map of Figure 2 is covered.

Phase 4 — The full film-region model (challenging — if time permits)**.** Replace the prescribed parabola with the coupled boundary-integral and lubrication scheme of §4: solve (10) and (11) together so the film pressure shapes the underside self-consistently. This lifts the rigid-parabola restriction while staying entirely within the film region, lets you predict the bounce map from a genuine deformation calculation, and is the natural bridge to the full Navier–Stokes simulations that motivate the whole enterprise. Build it first in planar (2D) geometry — which is how §4 sets it up — where the coupling is easiest to get right, and only then promote it to the axisymmetric case (§4.2, Appendix A).

A note on tools. The solvers here are small one-dimensional codes, and **Python** (with `numpy` and `scipy`) is an excellent choice — the finite-volume Reynolds solver of Exercise 5 is a few dozen lines, and the banded linear solve is a single library call. You are of course free to use whatever language you are comfortable with, but Python will get you to the first benchmark fastest and makes plotting and validation painless.

10 Reading

Start with the review, then dip into the primary sources as needed.

- **Sprittles**, *Gas microfilms in droplet dynamics: when do drops bounce?*, Annu. Rev. Fluid Mech. **56**, 91 (2024). The review to read first: it lays out the whole gas-film picture, including the gas-kinetic and van der Waals physics of Parts 2–3.

- **Hicks & Purvis**, *Air cushioning and bubble entrapment in three-dimensional droplet impacts*, J. Fluid Mech. **649**, 135 (2010). The starting point for this project: the film-region approach and the parabolic drop-surface assumption of §1.1.
- **Mandre, Mani & Brenner**, *Precursors to splashing of liquid droplets on a solid surface*, Phys. Rev. Lett. **102**, 134502 (2009), and **Mani, Mandre & Brenner**, *Events before droplet splashing on a solid surface*, J. Fluid Mech. **647**, 163 (2010). The same inviscid-liquid, harmonic-pressure gas-film approach as Hicks & Purvis, developed independently: they show the drop first drains (or compresses) the trapped gas and skates on it rather than contacting the solid, and analyse when that film eventually breaks down. Read these alongside Hicks & Purvis for the method and for why contact is not inevitable (§1).
- **Kolinski et al.**, *Skating on a film of air: drops impacting on a surface*, Phys. Rev. Lett. **108**, 074503 (2012). The experiments that directly image the trapped air film and its rupture on a solid.
- **de Ruiter et al.**, *Wettability-independent bouncing on flat surfaces mediated by thin air films*, Nat. Phys. **11**, 48 (2015). Beautiful experiments that measure the gas film directly and show the cushioned bounce; the images your solver output should be compared against.
- **Kolinski et al.**, *Drops can bounce from perfectly hydrophilic surfaces*, Europhys. Lett. **108**, 24001 (2014). A drop bouncing off a flat solid even when the solid would be perfectly wetted — direct evidence that the air film, not wettability, decides the outcome (§1).
- **Thoroddsen et al.**, *The air bubble entrapped under a drop impacting on a solid surface*, J. Fluid Mech. **545**, 203 (2005), and **Bouwhuis et al.**, *Maximal air bubble entrainment at liquid-drop impact*, Phys. Rev. Lett. **109**, 264501 (2012). Measurements of the trapped central bubble — the experimental counterpart of the dimple and its $\mathrm{St}^{-2/3}$ scaling (8) (§2.1).
- **Pan, Law & coworkers** (2008), experiments on colliding tetradecane drops showing the merge–bounce–merge sequence of Figure 2 as the Weber number increases — the outcome map the project targets.
- **Oron, Davis & Bankoff**, *Long-scale evolution of thin liquid films*, Rev. Mod. Phys. **69**, 931 (1997). Background on lubrication theory and the disjoining pressure; read §II for the derivation behind Exercise 1.

Supervision. The project is supervised by Prof. James Sprittles (Zeeman Building, office B2.11; `J.E.Sprittles@warwick.ac.uk`), with day-to-day co-supervision from Dr T. D. Dang. Come and talk through the exercises — that is what they are for.

A The axisymmetric formulation

The main text builds the coupled solver in planar (2D) geometry and then gives the axisymmetric governing equations in §4.2; this appendix records the *numerical* detail of that promotion — the quadrature bookkeeping that would otherwise clutter the first build.

Film equation. With the $1/r$ factor of (12), the finite-volume solver of §3 carries over directly: each face flux picks up the geometric weight $r_{i+1/2}$, the axis cell uses the regularity condition $\partial_r p = 0$, and $p \to 0$ closes the outer edge. The linear system stays tridiagonal.

Boundary integral. The axisymmetric relation (13) is obtained from the general Hicks & Purvis surface integral by doing the azimuthal integral analytically. Writing the source point as $\mathbf{x}' = (r'\cos\phi, r'\sin\phi)$ and the target on the axis line at radius r, the distance between them is $[r^2 + r'^2 - 2rr'\cos\phi]^{1/2}$, and the ϕ-integral of the Hicks & Purvis kernel over $[0, 2\pi)$ produces complete elliptic integrals of the first and second kind, $K(m)$ and $E(m)$, with modulus $m = 4rr'/(r + r')^2$. The result is a single radial kernel $\mathcal{K}(r, r')$ carrying a weight r' from the area element $r'\,\mathrm{d}r'\,\mathrm{d}\phi$. That kernel has an integrable (logarithmic) singularity as $r' \to r$, coming from $K(m) \to \infty$ at $m \to 1$; it is handled by the same desingularisation — subtract the target-point value of $\partial p/\partial r'$ — used in the planar Hilbert transform (10). In practice you tabulate $\mathcal{K}(r, r')$ once on the

radial grid (SciPy's `scipy.special.ellipk/ellipe` give K and E directly) and reuse it at every time step.

What does not change. The two-step iteration — solve the film equation for the pressure at fixed gap, then update the surface from the boundary integral, repeat to convergence — the time stepping, the release condition, and the validation strategy are all identical to the planar build. Promoting from 2D to axisymmetric is a change of geometry and quadrature, not of method.

B The finite-volume method, and how to code it

The Reynolds equation (6) is the workhorse of Part 1, and the **finite-volume** (FV) method is the natural way to discretise it. This appendix explains what the method is and turns it into a few lines of Python, so Exercise 5 is a matter of filling in a template rather than inventing a scheme.

The idea: conserve, cell by cell. A finite-volume scheme does not approximate derivatives at points (that is finite *differences*); instead it divides the domain into small **cells** and enforces the conservation law exactly on each one. Our equation, in planar form,

$$\frac{\partial}{\partial x}\left(h^3\,\frac{\partial p}{\partial x}\right) \;=\; \frac{12}{\mathrm{St}}\,\frac{\partial h}{\partial t},$$

is a balance: the divergence of a *flux* $q = -h^3\,\partial_x p$ equals a source $-(12/\mathrm{St})\,\partial_t h$. Integrate it over one cell $[x_{i-1/2}, x_{i+1/2}]$ of width Δx. The integral of the divergence is, exactly, the flux in at the left face minus the flux out at the right face (the divergence theorem in one dimension), so

$$q_{i-1/2} - q_{i+1/2} \;=\; \frac{12}{\mathrm{St}}\,\frac{\partial h_i}{\partial t}\,\Delta x,$$

where h_i is the average gap in cell i. Nothing has been approximated yet — this is just accounting. The scheme comes from writing each *face* flux in terms of the neighbouring cell values.

Face fluxes. The flux at the face between cells i and $i{+}1$ needs the pressure gradient there, which we approximate by a difference of the two cell pressures, and the conductance $K = h^3$ there, which we take as the value of h^3 interpolated to the face (an average of the two cells):

$$q_{i+1/2} \;=\; -\,K_{i+1/2}\,\frac{p_{i+1}-p_i}{\Delta x}, \qquad K_{i+1/2} = \tfrac{1}{2}\big(h_i^3 + h_{i+1}^3\big).$$

Substituting the two face fluxes into the cell balance gives, for every interior cell, a linear equation relating p_{i-1}, p_i and p_{i+1}. Collecting them is a **tridiagonal** system $\mathbf{A}\,\mathbf{p} = \mathbf{b}$: row i has $-K_{i-1/2}$ and $-K_{i+1/2}$ off the diagonal, their sum on the diagonal, and the source $-(12/\mathrm{St})\,\dot{h}_i\,\Delta x^2$ on the right (the minus sign is because we have negated the row to put the positive conductances on the diagonal). Symmetry ($q = 0$, i.e. $\partial_x p = 0$) closes the axis by dropping the left-face flux of the first cell; $p = 0$ closes the outer edge, imposed by reflecting the last cell in a ghost value $p_M = -p_{M-1}$.

Why FV and not FD here. Because the face-flux out of one cell is *identically* the flux into its neighbour, mass is conserved to machine precision no matter how coarse the grid — there is no spurious gas created or destroyed. That property, and the graceful way the variable conductance h^3 sits on the faces, is why FV is the standard choice for lubrication and conservation-law problems.

In Python. With `numpy` for the assembly and `scipy.linalg.solve_banded` (or `scipy.sparse`) for the solve, the pressure step is short:

```
import numpy as np
from scipy.linalg import solve_banded

def pressure_step(h, hdot, dx, St):
    # Cell-centred finite volume for  d/dx(h^3 dp/dx) = (12/St) dh/dt.
    # h, hdot: gap and its time-derivative at the M cell centres.
    # BCs: zero flux (symmetry) at the axis x=0; p=0 at the outer edge.
    M = h.size
    K = h**3
    Kf = 0.5*(K[:-1] + K[1:])          # conductance on the M-1 interior faces
    lo = np.zeros(M); di = np.zeros(M); up = np.zeros(M)
    b  = -(12.0/St)*hdot*dx*dx         # negated source (note the sign)
    # interior cells: -K_{i-1/2} p_{i-1} + (K_{i-1/2}+K_{i+1/2}) p_i
    #                                    - K_{i+1/2} p_{i+1} = b_i
    di[1:-1] = Kf[:-1] + Kf[1:]
    lo[1:-1] = -Kf[:-1]
    up[1:-1] = -Kf[1:]
    # axis cell 0: left face carries no flux, so only the right face acts
    di[0], up[0] = Kf[0], -Kf[0]
    # outer cell M-1: p=0 at the edge, imposed by an image ghost p_M = -p_{M-1},
    # which turns the right-face conductance into 2*K[-1]
    di[-1], lo[-1] = Kf[-1] + 2.0*K[-1], -Kf[-1]
    ab = np.zeros((3, M))              # banded storage for solve_banded
    ab[0,1:] = up[:-1]; ab[1,:] = di; ab[2,:-1] = lo[1:]
    return solve_banded((1,1), ab, b)
```

That is the whole solver core. To go axisymmetric, multiply each face flux by the face radius $r_{i+1/2}$ and each source by the cell radius r_i — two extra factors, as Appendix A describes, and nothing else changes.

How to trust it. Check the code against something you can compute by hand before you believe any bounce. Two independent tests: (i) freeze the gap as the parabola (2) and confirm the pressure matches the analytic squeeze-film result (7); (ii) the **method of manufactured solutions** — pick a smooth $p(x)$ you like, substitute it into the equation to find the source it requires, feed that source to your code, and confirm it recovers your chosen p to second order as you refine Δx. If the error falls by four when you halve Δx, the scheme is second-order and correct.